14.2.3 Alternative Forms of DNA

DNA is a remarkably flexible molecule. Considerable rotation is possible around a number of bonds in sugar-phosphate backbone and thermal fluctuations can produce bending, stretching and unpairing of the strands. Watson and Crick model of DNA is called as B-DNA or B-form. However DNA can exist in A or Z form. In 1979 Alexander Rich discovered Z form (Figure 14.10). Recently, several alternative forms of DNA have been discovered C-form, D-form and E-form. The B-form of DNA is the most stable structure and is therefore the standard point of reference in any study of the properties of DNA (Table 14.1).

Table 14.1: Properties of different forms of DNA

	A form	B form	Z form
Helical sense	Right handed	Right handed	Left handed
Diameter	~ 26A°	~ 20 A°	~ 18 A°
Base pairs per helical turn	11	10	12
Distance between adjacent bases	2.6 A°	3.4 A°	3.7 A°



Bacterial genome size

- Bacterial genomes are typically expressed in Mb
- The length of Bacterial genomes are typically in the mm range and therefore 1000X bigger than the typical bacterial size.
- The mass of Bacterial genomes are typically in 10^{-3} pg(picogram) range.

Conversions

1 Kb = 10^3 bp (base pairs)

1 Mb = 10^6 bp

1 Gb = 10^9 bp

1 bp ≈ 0.33nm

1 kb ≈ 0.33μm

1 Mb ≈ 0.33mm

1 Gb ≈ 0.33m

1 pg = 10^{-12} g

1pg = 978 Mb

Number of base pairs = mass in pg × (0.978 × 10^9)

1 kb ≈ 10^{-6} pg

1 Mb ≈ 10^{-3} pg

1 Gb ≈ 1 pg

	Bacteria	Virus, organelles	Eukaryotes
Genomes	Small(Mb)	Tiny (Kb)	Large (Gb)
Gene Density	High	High	Low
Example	*E.coli* 5000 genes	Bacteriophages 10-100 genes	*Homo sapiens* 25000 genes

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